

DEEPAK KUMAR, Ph.D.

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Professional Summary

Structural Engineer with a Ph.D. in Civil Engineering and a track record of applying computer vision, deep learning, and data-driven modeling to bridge inspection, deterioration forecasting, and infrastructure monitoring. Experience spans vision-based hazard detection, multimodal acoustic sensing, survival-analysis deterioration modeling, and LLM-based workflows for structured data extraction and automated decision support. 2+ years of university-level teaching experience, including graduate instruction, student assessment, and curriculum support, with ability to teach engineering mechanics, statistics, dynamics, and finite element analysis courses using modern educational technologies. Active researcher with publications in peer-reviewed journals and conferences.

Education

Ph.D. & M.Phil. in Civil Engineering (Structures) August 2019 - December 2024

The City College of New York, CUNY

Dissertation: *Data Driven Bridge Deck Deterioration Modeling and Maintenance Intervention Scheduling*

Advisor: Dr. Anil K. Agrawal, *Ph.D., P.E.*

B.Tech in Civil Engineering (Honors.)

July 2015 - May 2019

Indian Institute of Technology, Varanasi

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Relevant Graduate Courses : Finite Element Methods in Engineering, Applied Stress Analysis, Advanced Strength of Materials, Bridge Engineering, Structural Dynamics, High-Rise Building Design and Analysis, Analytical Methods CE

AI/ML Courses for Engineering Application: Artificial Intelligence for Engineering Applications, Introduction to Data Science, Database Systems, Generative AI for Natural Language Processing.

Teaching Experience

Role: Adjunct Assistant Professor

The City College of New York of CUNY

January 2025 - May 2025

- Independently taught graduate course on structural dynamics (CE I3000), managed lectures, prepared course contents, and MATLAB/Python-based projects focused on dynamic analysis of bridges and buildings.

Role: Teaching Assistant

The City College of New York of CUNY

August 2021 - December 2024

- Assisted senior students in lab sessions (SAP2000), providing support with assignments, quizzes, projects, and holding office hours for courses CE 435 (Dynamics of Civil Engineering Systems), CE 401 (Review of Civil Engineering Fundamentals) & CE 440 (Finite Element Analysis of Structures).

Publications & Presentations

- Invited Speaker, AI Applications for Intelligent Bridge Infrastructure Management**
ASCE SEI Metropolitan Section & NYSED Sponsor #6 Webinar (1 PDH), December 2025 ([webinar](#))
- Kumar, D., Agrawal, A. K. (2026). **AFT-Weibull Modeling of Bridge Deck Deterioration with Optimized Censoring**. *Journal of Bridge Engineering*, 31(9), 04026046. ([DOI](#))
- Kumar, D., Agrawal, A. K. (2025). **Smart Acoustic Sounding for Automated Delamination Detection in Concrete Bridge Decks**. *International Journal of Bridge Engineering, Management and Research*, 2(4), 214250033-1:214250033. ([DOI](#))
- Kumar, D., Agrawal, A. (2025). **Advancing Bridge Infrastructure Management through Artificial Intelligence: A Comprehensive Review**. *International Journal of Bridge Engineering, Management and Research*, 2(3), 214250021-1:18. ([DOI](#))
- Kumar, D. (2024). **Data Driven Bridge Deck Deterioration Modeling and Maintenance Intervention Scheduling** (Doctoral dissertation, The City College of New York).
- Agrawal, A. K., Kumar, D. (2024). **Smart Sounding System for Autonomous Evaluation of Concrete and Metallic Structures**.
- Agrawal, A., Kumar, D. (2023). **An Artificial Intelligence (AI) Based Overheight Vehicle Warning System for Bridges**.
- Kumar, D., Agrawal, A.K., Cao, R., Zhan, L., Wei, J. (2023). **Damage Detection in Concrete Slab Using Smart Sounding**. In: Rizzo, P., Milazzo, A. (eds) **European Workshop on Structural Health Monitoring**. EWSHM 2022. Lecture Notes in Civil Engineering, vol 254. Springer, Cham. ([DOI](#))

Research Experience

Role: Graduate Research Associate

Research Foundation of City University of New York (RFCUNY)

August 2019 - December 2024

♦ Smart Sounding System for Autonomous Evaluation of Concrete Structures

INSPIRE-University Transportation Center (INSPIRE-UTC)

August 2019 - June 2024

- Designed and built a low-cost smart sensing device integrating MEMS microphones, Raspberry Pi, and a Bluetooth acoustic source to detect delamination in concrete decks.
- Programmed a Python-based system for data acquisition, advanced signal processing, and **real-time visualization, deployed and tested on laboratory and field specimens**.
- Conducted experimental validation by comparing device performance with traditional chain-drag and hammer methods, advancing portable **structural health monitoring** and sensing technologies.

♦ American Association of State Highway and Transportation Officials (AASHTO) and National Bridge Inventory (NBI) Element Deterioration Rates for Bridge Management System

Connected Cities with Smart (C2Smart) Transportation Center

October 2020 - December 2022

- Performed SQL/VBA queries for **developing macros to automate** the data cleaning, analysis and classification of different elements present in bridge inventory database having about 20,000 bridge inspection records.
- Designed **computational models** for bridge deck deterioration using Markov Chain, Weibull survival analysis and Accelerated Failure Time (AFT) models.
- Built a MATLAB desktop platform with 60+ features for deterioration prediction, data visualization, and model updating, providing agencies with a decision-support tool for budgeting and intervention planning, with built-in features to **support for future database update**.

♦ An Artificial Intelligence (AI) Based Overheight Vehicle Warning System for Bridges

Connected Cities with Smart (C2Smart) Transportation Center

February 2022 - August 2023

- A Developed a **machine learning**-powered vision system for overheight vehicle detection, integrating YOLOv8, vanishing point estimation, and 3D bounding box generation for **real-time hazard detection**.
- The module provides cost-effective installation on bridges for **continuous monitoring** and warning to prevent bridge strikes by overheight trucks.

Role: Research Intern

Seismic Displacement Demand of Curved Bridge in OpenSees

The City College of New York (CUNY) - Online

March 2018 - November 2018

- Modeled a 3DOF short span curved bridge in OpenSees to analyze its seismic response, incorporating **non-linear soil behavior** and dynamic characteristics.

Behaviour of Steel Plates in Fire

Indian Institute of Technology, Hyderabad, India

May 2018 - July 2018

- Conducted a parametric study in ABAQUS to investigate the effects of slenderness, temperature, imperfections, boundary condition, and materials using Eurocode and NIST under fire conditions.

Industrial Experience

Role: Structural/Bridge Engineer

ATANE Engineers, Architects and Land Surveyors, DPC

November 2024 - Present

Responsibilities include conducting structural analyses of steel and concrete structures, including complex bridges, in accordance with AASHTO guidelines. Additionally, model bridges of varying complexities, review bridge inspection reports, and collect deterioration data for steel and concrete members.

- ♦ Performed Level 2 load rating analyses in compliance with NYSDOT Region 11 protocols using AASHTOWare BrR. Reviewed as-built and contract plans; updated and validated bridge models with latest inspection data, section loss, and field conditions. Evaluated HL-93, HS-20, H-20, EV2/EV3 loadings to assess structural adequacy and determine posting requirements.
- ♦ Completed NJDOT-assigned load rating of Structure, a curved steel plate girder bridge. Developed a 3D FEM model in Midas Civil, in compliance with AASHTO and NJDOT standards, for multiple vehicle types including HL-93, HS-20, SHVs, and FAST Act Emergency Vehicles.
- ♦ Rated bridges using AASHTOWare BrR, applying LRFR with HL-93 and LFR with HS-20, H-20, and FAST Act Emergency Vehicles. Assessed section loss, impact-related damage, and other deficiencies to support accurate Level 2 load rating evaluations and ensure compliance with AASHTO standards.

Technical Skills & Interests

- **Programming Languages:** Python, MATLAB, SQL
- **Computational Modeling Tool:** SAP2000, STAAD-Pro, ABAQUS, AASHTOWare BrR, Midas Civil, OpenSees, ETABS, AutoCAD

Relevant Projects

- **Grillage Analysis of Two Span Continuous Bridge:** Performed grillage analysis and sensitivity studies on structural parameters to compare bridge girder responses with AASHTO load distribution methods.
- **Stress Concentration Factor for Rectangular Filleted Bar:** Investigated stress concentration in a Rectangular Filleted Plate using FEM, focusing on geometric discontinuities.
- **High Rise Building Design and Analysis:** This project involved designing and analyzing a 10-story and a 60-story building using ETABS, considering seismic and wind loads per IBC 2018 and ASCE 7-16 codes, with detailed modeling, member design, and schematic drawing outputs.
- **Modifying the Given FEM Program to Include the Constant Strain Triangular Element:** Modified MATLAB's FEM code, originally supporting only quadrilateral elements, to include triangular elements, achieving consistent results with STAAD-Pro for stress concentration in a rectangular plate with grooves.
- **Response History and Modal Analysis under El-Centro Earthquake:** Performed response history analysis of an SDOF system under the El-Centro earthquake to determine maximum response and conducted modal analysis for a shear building using the Stodola method to calculate natural frequencies, mode shapes, and response.
- **One-Way and Two-Way Slab Systems Design:** Designed and analyzed both one-way and two-way slab systems, calculating material quantities and comparing the relative costs of each system per square foot.
- **Two-Span Continuous Girder Design:** Designed a two-span continuous girder with bearing stiffeners and connections, including moment and shear envelopes, material specifications, and a detailed sketch.
- **Severstal Steel Defect Detection (Kaggle):** Performed defect detection in Steel bars using simple Convolution Neural Networks and transfer learning with ResNet model, & applied segmentation for defect regions in image.

Certificate Courses & Webinars

- NYSDOT- PDH Course DOT25003 - **Bridge Load Rating, Posting, and Overweight Permitting** (2025)
- ASCE - AW1092618 - **Practical Life-Cycle Analysis for Bridges** (2025)
- NYSDOT – PDH Course DOT25034 - Load Rating of a Simple Span Composite Steel Stringer Bridge (2025)
- Virtual Symposium on Protection of Bridge Piers from Large Ship Impacts (August 7, 2024) [[Certificate](#)]
- Risk Based Tolerant Seismic Design of Structure: A New Paradigm by Prof. Achintya Haldar, University of Arizona, USA (18 Dec-27 Dec, 2017) [[Reference](#)]
- Recent Advances in Structural Dynamics for Protection of Bridges Against Extreme Hazards, under GIAN Scheme, MHRD Govt. of India (01 Nov-05 Nov, 2017) [[Reference](#)]
- Advanced Techniques for Traffic Data Analysis, Visualization and State Estimation for Indian Cities (20-24 Dec, 2021)
- Generative AI Applications with LangChain (Feb 2024)([Great Learning Portfolio](#))

Leaderships & Management

- Organized an “Introduction to Deep Learning with MATLAB” workshop for high school students and supervised interns on image detection and segmentation projects using Python.
- Conducted exams, laboratory sessions, and workshops for cohorts of 50+ students, ensuring effective knowledge transfer and engagement.
- Mentored summer interns, senior undergraduates, and master's students on independent research projects in structural modeling, AI applications, and civil infrastructure systems at CCNY.
- Trained junior engineers at ATANE in company standards, including technical report preparation, structural modeling workflows, and load rating calculations.
- Coordinated with bridge inspection team leaders and state transportation agencies to review safety issues, evaluate field data, and recommend flag posting requirements.

I hereby declare that all the information provided herein is true and accurate to the best of my knowledge.

Dr. DEEPAK KUMAR